

Portfolio Analysis and Optimisation of a Diversified U.S. Equity Portfolio

An Application of Modern Portfolio Theory

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Abstract

This report analyses a diversified portfolio of 30 U.S. equities across seven sectors (technology, financials, healthcare, consumer goods, automobiles, energy and industrials) from January 2020 to January 2025, using Modern Portfolio Theory. An equal weight portfolio, with each stock at 3.33%, delivered an expected annual return of 23.78% at 21.40% annualised volatility, a Sharpe Ratio of 0.92 and a maximum drawdown of 33.24% during the COVID 19 shock of early 2020.

A Monte Carlo simulation of 10,000 random portfolios was used to approximate the Efficient Frontier. The Maximum Sharpe Portfolio returned 29.53% at 22.60% volatility, a Sharpe Ratio of 1.13. The Minimum Volatility Portfolio returned 20.44% at 18.55% volatility, a Sharpe Ratio of 0.89. Both hold the same 30 stocks as the benchmark, so the improvement comes from allocation alone, mainly a shift toward Apple, Exxon Mobil, Costco, Meta Platforms and Amazon. The report also sets out the limits of the approach, including in sample optimisation and unstable correlations.

Keywords: Modern Portfolio Theory, portfolio optimisation, Monte Carlo simulation, Efficient Frontier, Sharpe Ratio, diversification, risk adjusted return

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1. Introduction and Objectives

Anyone building a portfolio of risky assets has to balance expected return against risk. Modern Portfolio Theory (Markowitz, 1952) gives a framework for measuring and optimising that balance. This report applies it to 30 U.S. listed equities across seven sectors, using daily closing prices from 1 January 2020 to 1 January 2025 and a Python workflow that runs from data download to portfolio optimisation. All calculations, figures and outputs come from the author's original Jupyter notebook. The outputs have also been prepared for an interactive Streamlit dashboard, which is a presentation layer and not part of the analysis.

The aim is to build, evaluate and optimise a diversified equity portfolio using Modern Portfolio Theory. The study:

- downloads and validates historical price data and calculates daily and cumulative returns for each stock;
- builds an equal weight benchmark and measures its return, volatility, Sharpe Ratio, rolling volatility and maximum drawdown;
- analyses the covariance and correlation of the constituents;
- runs a Monte Carlo simulation to approximate the Efficient Frontier and identify the Maximum Sharpe and Minimum Volatility portfolios.

Scope. The portfolios are long only and hold public equities only. The study excludes transaction costs, taxes, rebalancing, short selling, out of sample testing and other asset classes. All results are historical and do not predict future returns.

2. Literature Review

Modern Portfolio Theory holds that a rational investor should seek the highest return for a given level of risk, and should judge an asset by its contribution to the whole portfolio, not on its own (Markowitz, 1952). Its main contribution is a formal basis for diversification. Assets that do not move in lockstep, measured by covariance and correlation (from minus one to plus one), lower portfolio volatility without a proportional loss of return, and the lower the correlation, the larger the benefit.

Risk adjusted performance is usually measured by the Sharpe Ratio, the return above the risk free rate divided by volatility (Sharpe, 1966). The Efficient Frontier is the set of portfolios with the highest expected return at each level of risk, and any portfolio below it is inefficient. Two points on it matter here: the Maximum Sharpe Portfolio, with the best risk adjusted return, and the Minimum Volatility Portfolio, with the lowest risk. With many assets the exact frontier needs a constrained quadratic optimisation, so it is often approximated by simulating a large number of randomly weighted portfolios, which is the approach used in this report.

The mean variance framework has known weaknesses. Expected returns, variances and covariances must all be estimated from historical data, and expected returns carry far more estimation error than volatility or correlation, so optimal weights can swing on small changes in inputs. The theory also assumes normally distributed returns, stable correlations and costless trading, none of which holds fully in real markets. These points are taken up in Section 6.

3. Methodology

Expected portfolio return is the dot product of the weight vector w and the vector of mean daily returns μ , annualised with 252 trading days:

$$E[R_p] = (w^T \mu) \times 252$$

Portfolio volatility uses the covariance matrix Σ of daily returns, which captures each stock's own variance and how the stocks move together, annualised with $\sqrt{252}$:

$$\sigma_p = \sqrt{(w^T \Sigma w) \times 252}$$

The Sharpe Ratio uses a constant annual risk free rate R_f of 4%, roughly the yield on short term U.S. Treasuries:

$$Sharpe = (E[R_p] \text{ minus } R_f) / \sigma_p$$

Rolling volatility is the annualised standard deviation of portfolio returns over a moving 30 day window, which reacts to short term shifts while staying smooth enough to show sustained periods of high risk. Maximum drawdown is the largest percentage fall from the running peak of cumulative portfolio value:

$$Drawdown(t) = (Cumulative(t) \text{ minus } RunningMax(t)) / RunningMax(t)$$

The Efficient Frontier was approximated with 10,000 randomly weighted portfolios, with weights summing to 100% and no short selling. Expected return, volatility and Sharpe Ratio were calculated for each one. The highest Sharpe Ratio in the set defines the Maximum Sharpe Portfolio and the lowest volatility defines the Minimum Volatility Portfolio. A fixed seed (`np.random.seed(18)`) makes the simulation reproducible.

4. Data and Portfolio Construction

Daily prices were downloaded with the `yfinance` library (Yahoo Finance) for 1 January 2020 to 1 January 2025 on an adjusted close basis (`auto_adjust=True`), so dividends and splits are reflected in returns. The data were checked for structure, dimensions and missing values. Daily returns were calculated with `pct_change()` and the undefined first row was dropped. Working in returns puts every stock on the same scale and feeds all later calculations.

The 30 stocks span seven sectors to limit concentration risk and spread exposure across drivers such as interest rates, consumer spending, commodity prices and business investment. Technology has the most stocks. Amazon and Alphabet are classified as Technology, as in the notebook, although standard taxonomies place them in consumer discretionary and communication services.

Table 4.1: Portfolio Constituents by Sector

Sector	Constituent Tickers	No. of Stocks
Technology	AAPL, MSFT, NVDA, AMZN, GOOG, META, AMD, ORCL	8
Financials	JPM, BAC, GS, V, MA	5
Healthcare	JNJ, PFE, LLY, MRK, ABBV	5
Consumer	WMT, COST, PG, KO, MCD	5
Automobiles	TSLA, GM, F	3
Energy	XOM, CVX	2
Industrials	CAT, GE	2

The benchmark is an equal weight portfolio, with each stock at 1/30 (about 3.33%), confirmed to sum to 100%. It is a neutral baseline that assumes nothing about which stocks are more attractive, so any improvement from optimisation comes from reweighting alone.

5. Results

5.1 Correlation Structure

Covariance and correlation matrices were computed for all 30 stocks, and the full matrices are in the notebook. Correlations are positive throughout and highest within sectors (Figure 5.1). The technology names (Apple, Microsoft, NVIDIA, Alphabet, Amazon, AMD) form a clear cluster, as do the banks (JPMorgan, Bank of America, Goldman Sachs) and the energy pair (Chevron, Exxon Mobil). Healthcare and consumer staples names such as Johnson & Johnson, Pfizer, Procter & Gamble and Walmart correlate more weakly with the technology cluster. There are no negatively correlated pairs, so the diversification benefit comes from the varying degree of positive correlation across sectors.

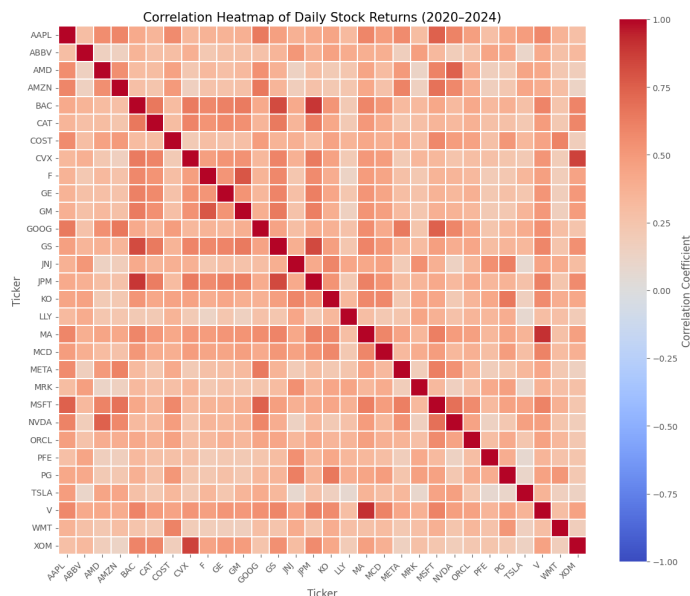


Figure 5.1: Correlation Heatmap of Daily Stock Returns (2020 to 2024)

5.2 Returns and Growth

NVIDIA had the highest average daily return (0.3050%), followed by Tesla (0.2997%) and Eli Lilly (0.1663%). Pfizer was lowest at 0.0052%, with Johnson & Johnson and Coca Cola close behind. Growth oriented technology and healthcare names sit well above the defensive stocks (Figure 5.2).

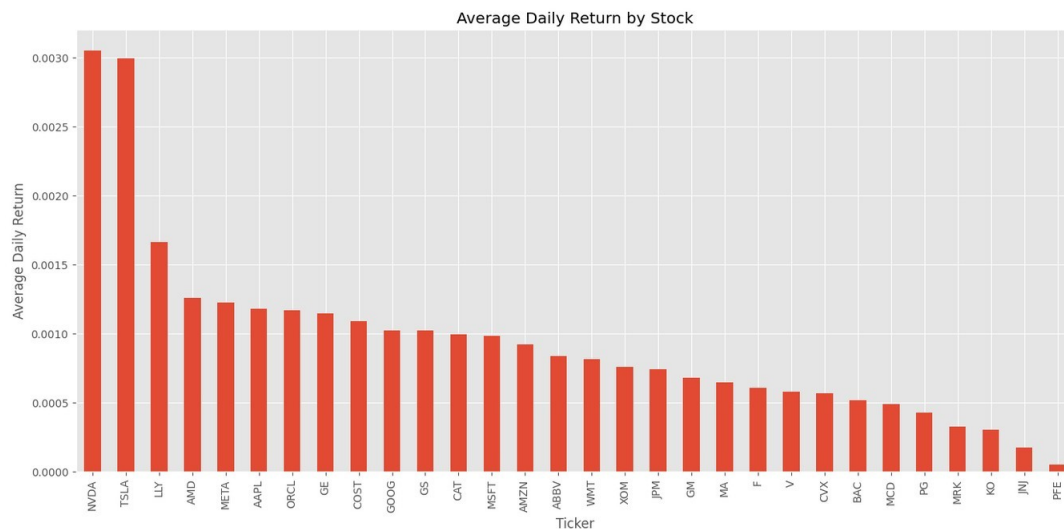


Figure 5.2: Average Daily Return by Stock

By cumulative return, the five strongest performers were NVIDIA, Tesla, Eli Lilly, Apple and Costco (Figure 5.3). This ranking differs from average daily return because compounded returns are always at or below the arithmetic average, and the gap widens as volatility rises. Tesla, with large daily swings, gives up more to this volatility drag than a steadier stock such as Costco. NVIDIA still finished well ahead despite sharp swings in 2022 and 2023.

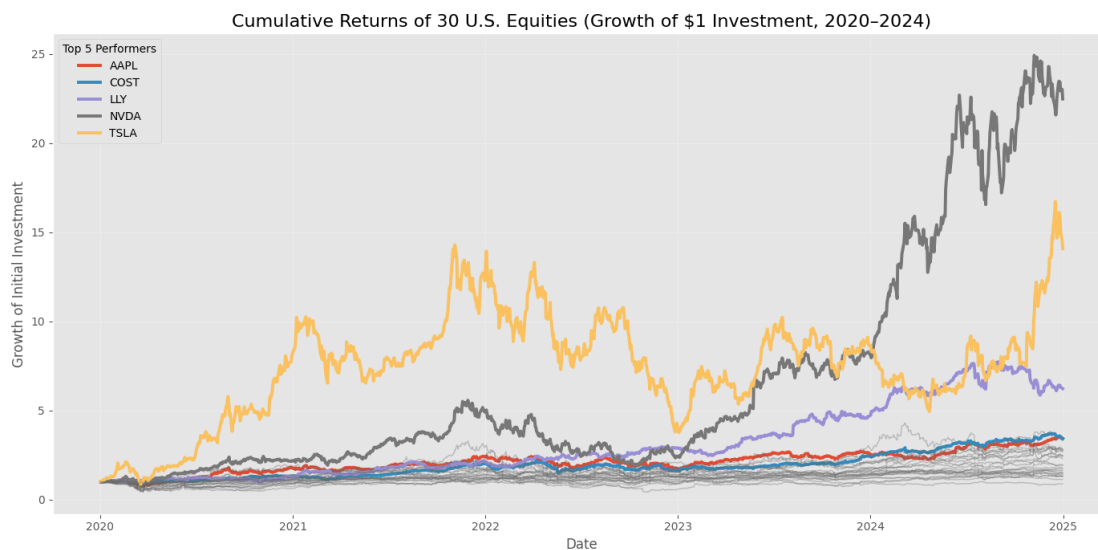


Figure 5.3: Cumulative Returns of 30 U.S. Equities (Growth of a \$1 Investment, 2020 to 2024)

5.3 Risk Analysis

The equal weight portfolio had daily volatility of about 1.35%, or 21.40% annualised, against an expected annual return of 23.78%. Exposure to volatile growth names such as NVIDIA, Tesla and Meta drives this level, partly offset by diversification, which kept portfolio risk well below that of its most volatile holdings.

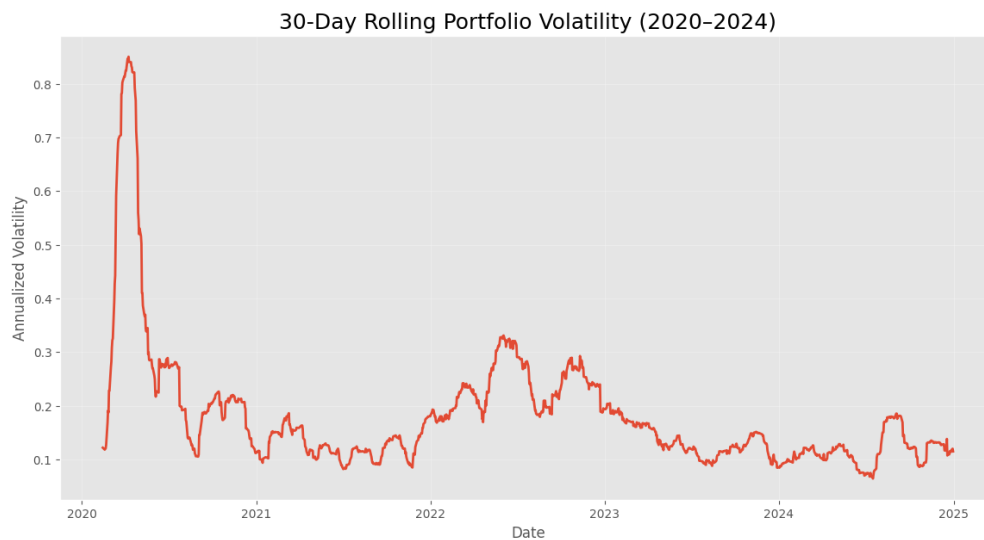


Figure 5.4: 30 Day Rolling Portfolio Volatility (2020 to 2024)

Rolling volatility averaged 18.10% but peaked at 85.03% in early 2020 as COVID 19 hit, then fell to a low of 6.42%. A 90 day window showed the same spike, so it reflects a real market event and not the window choice. Portfolio risk is not fixed: it rises sharply under stress and eases once conditions normalise.

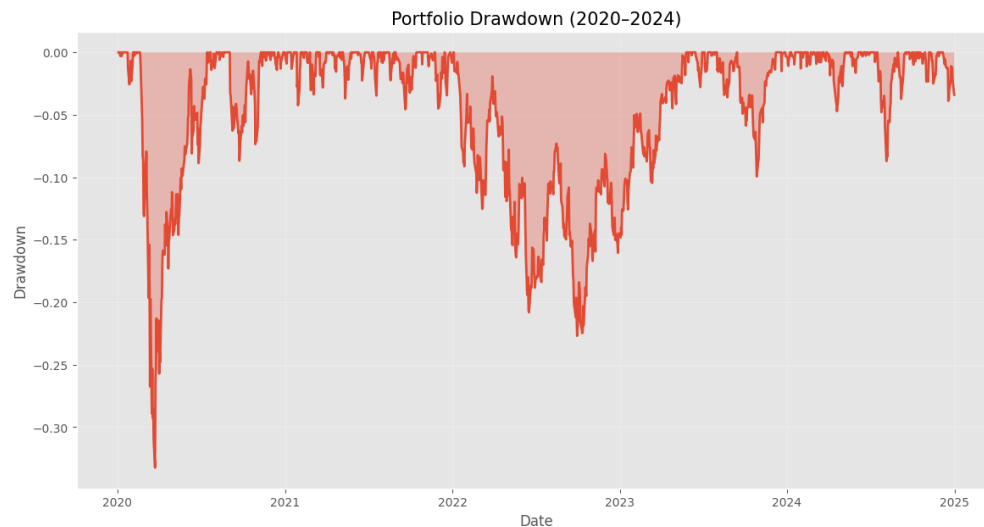


Figure 5.5: Portfolio Drawdown (2020 to 2024)

Maximum drawdown was 33.24%, so an investor who bought at the previous peak would have seen the holding fall by about a third before it recovered. That is consistent with a growth tilted portfolio over a sample that includes the COVID 19 crash, the 2022 bear market and rapidly rising interest rates, and the portfolio repeatedly climbed back to new highs. Drawdown measures something different from volatility. It is the worst case for an investor who has to sell at the trough, which is sequence of returns risk. An investor who stayed invested recovered the loss within the sample, but one forced to sell at the bottom would not.

With a 4% risk free rate, the Sharpe Ratio was 0.92, just under the commonly cited level of 1.0. The allocation paid investors reasonably for the risk taken but was not the most efficient available, which Section 5.4 tests.

5.4 Portfolio Optimisation

Figure 5.6 shows the 10,000 simulated portfolios. Return rises with volatility, as the theory predicts. Figure 5.7 marks the Maximum Sharpe and Minimum Volatility portfolios, and Table 5.1 compares them with the benchmark.

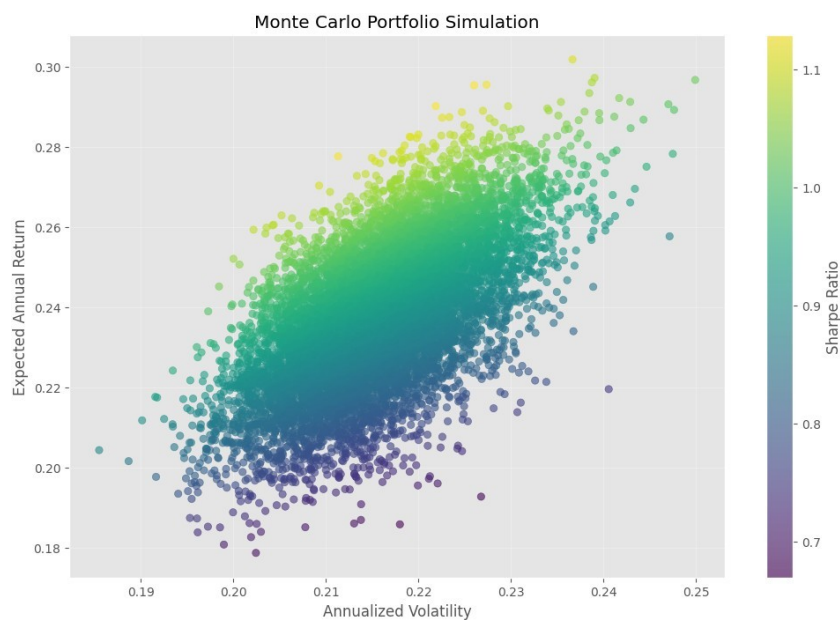


Figure 5.6: Monte Carlo Portfolio Simulation (10,000 Portfolios)

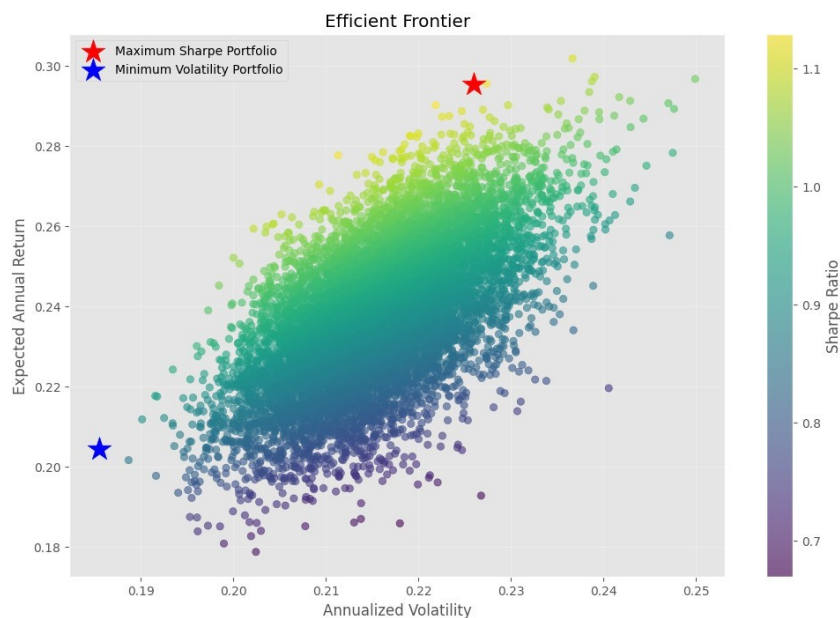


Figure 5.7: Efficient Frontier with Maximum Sharpe and Minimum Volatility Portfolios

Table 5.1: Optimised Portfolios Compared with the Equal Weight Benchmark

Portfolio	Expected Return	Annualised Volatility	Sharpe Ratio
Equal Weight Portfolio	23.78%	21.40%	0.92
Maximum Sharpe Portfolio	29.53%	22.60%	1.13
Minimum Volatility Portfolio	20.44%	18.55%	0.89

The Maximum Sharpe Portfolio adds 5.75 percentage points of expected return for only 1.20 points more volatility, so the gain comes mainly from better allocation and not from extra risk. The Minimum Volatility Portfolio cuts volatility by 2.85 points but gives up 3.34 points of return, and its Sharpe Ratio of 0.89 sits marginally below the benchmark. Minimising risk alone does not improve risk adjusted efficiency.

5.5 Portfolio Allocation

The Maximum Sharpe Portfolio gives larger weights to assets that contribute more to risk adjusted return. The five largest holdings (Table 5.2) account for about 35.5% of capital, and Figure 5.8 shows the top ten.

Table 5.2: Largest Allocations in the Maximum Sharpe Portfolio

Rank	Ticker	Company	Portfolio Weight
1	AAPL	Apple Inc.	7.33%
2	XOM	Exxon Mobil Corporation	7.26%
3	COST	Costco Wholesale Corporation	7.13%
4	META	Meta Platforms, Inc.	7.11%
5	AMZN	Amazon.com, Inc.	6.67%

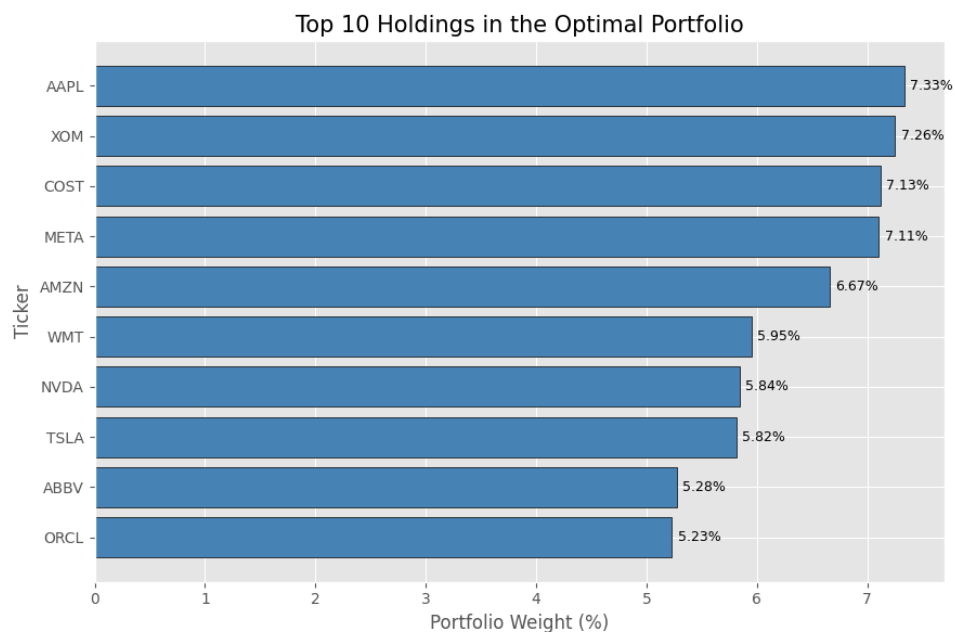


Figure 5.8: Top 10 Holdings in the Optimal Portfolio

The largest positions span Technology (Apple, Meta, Amazon), Energy (Exxon Mobil) and Consumer (Costco), and no stock dominates. The improvement therefore comes from weighting the existing diversification more efficiently, in line with the correlation patterns in Section 5.1, and not from concentrating in correlated stocks. Exxon Mobil ranks second despite a modest average return, which shows the optimiser rewards diversification value as well as return. Its weak correlation with the technology core lowers volatility in a way a return ranking would miss.

5.6 Strategy Comparison

Table 5.3: Summary Comparison of Portfolio Strategies

Portfolio	Expected Return	Annualised Volatility	Sharpe Ratio	Change in Sharpe Ratio vs Equal Weight
Equal Weight Portfolio	23.78%	21.40%	0.92	Benchmark
Maximum Sharpe Portfolio	29.53%	22.60%	1.13	+0.21 (+22.8%)
Minimum Volatility Portfolio	20.44%	18.55%	0.89	Down 0.03 (down 3.3%)

Reweightings alone lifted the Sharpe Ratio by about 23% over the benchmark, with the same 30 stocks.

6. Discussion

6.1 Allocation Matters More Than Selection

The equal weight and Maximum Sharpe portfolios hold the same 30 stocks, so the rise in Sharpe Ratio from 0.92 to 1.13 comes entirely from reweighting. Equal weighting is a common default, and here it left efficiency unused. The optimiser also did not simply back past winners: NVIDIA and Tesla, the two strongest individual stocks, are not among the five largest holdings. A stock's contribution depends on its own return and volatility and on how it moves relative to every other holding, not on its standalone performance.

6.2 Maximum Sharpe Versus Minimum Volatility

The two portfolios serve different objectives. Near the low risk end of the frontier the return profile is flatter, so further cuts in volatility cost more return than they save in risk. A pension fund close to its payout phase may accept the lower Sharpe Ratio of the Minimum Volatility Portfolio for its smaller swings, while an investor with a longer horizon would likely prefer the Maximum Sharpe balance. The frontier is a map of efficient choices, and the right point on it depends on the investor.

6.3 Diversification and COVID 19 Stress

With no negatively correlated pairs, diversification here is a matter of degree. All 30 stocks share exposure to monetary policy, growth and investor sentiment, so the benefit is a within asset class effect: a better Sharpe Ratio, not offsetting positions. Stronger diversification would probably need other asset classes such as government bonds. Reweighting still kept the portfolio broad, since the top five holdings make up only 35.5% of capital.

The 85.03% volatility spike and the 33.24% drawdown both came with the early 2020 shock. Sector diversification lowered average risk over the five years but could not protect against a systemic event. Correlations between risk assets tend to rise in a sell off, and because the correlation matrix covers the full sample, it blends crisis and calm periods and probably overstates the protection available during the crisis. The portfolio did recover to new highs within the window, so the drawdown was a temporary peak to trough decline and not a permanent loss.

6.4 Limitations

Optimisation and evaluation use the same historical data, so the Sharpe Ratio improvement carries in sample bias and has not been tested on unseen data. Expected returns are historical averages, estimated with far more error than volatility or correlation, which makes optimal weights sensitive to inputs. Correlations are assumed stable but rise in crises. Transaction costs, taxes and rebalancing are excluded, and they would reduce the advantage of the Maximum Sharpe Portfolio over the benchmark, because an optimal portfolio drifts away from its weights as prices move. The risk free rate was held at 4% although short term rates moved considerably between 2020 and 2025, so the Sharpe Ratios are approximate.

6.5 Future Work and Practical Implications

The following extensions are proposed and were not performed in this study: out of sample or rolling window testing of the optimised weights; transaction costs with a rebalancing schedule; the Black Litterman model to reduce sensitivity to expected return estimates; factor analysis (size, value, momentum) to separate diversification from common factor exposure; and broader asset classes such as government bonds.

For portfolio construction, reweighting within a fixed universe improved risk adjusted return, so periodic reoptimisation is worth doing. The choice between a Maximum Sharpe and a Minimum Volatility orientation should follow the investor's objective and horizon. Correlation aware allocation finds efficient holdings that a return ranking would miss. Optimal weights are best treated as a long run guide and not as protection against a systemic shock, and historical optimisation should be paired with stress testing against past crisis periods.

7. Conclusion

This report applied Modern Portfolio Theory to 30 U.S. equities across seven sectors from January 2020 to January 2025, using return and correlation analysis, risk measurement and Monte Carlo optimisation. The equal weight portfolio returned an expected 23.78% at 21.40% volatility, a Sharpe Ratio of 0.92. Reweighting the same stocks produced a Maximum Sharpe Portfolio with a Sharpe Ratio of 1.13, while the Minimum Volatility Portfolio cut volatility to 18.55% at a Sharpe Ratio of 0.89. Allocation, not stock selection, drove the gain. The diversification available came from varying positive correlations across sectors, and it lowered average risk but did not prevent a 33.24% drawdown in the COVID 19 shock.

The main limitation is that every result is in sample, built on historical estimates, a constant 4% risk free rate and no trading costs. The project delivers a complete, reproducible Python workflow from data collection to optimisation, with the theory's assumptions checked against real market behaviour. Disciplined, correlation aware allocation can improve risk adjusted outcomes without changing what a portfolio holds, but its weights need out of sample validation before they are relied on.

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Appendix

A.1 Workflow and Tools

The workflow ran in this order: download historical prices, data validation, return calculations, portfolio statistics, risk analysis, Monte Carlo simulation, Efficient Frontier, Portfolio Analysis Report. Tools used were yfinance (market data), pandas (data manipulation), NumPy (numerical computing and matrix operations), Matplotlib (Figures 5.2 to 5.8), Seaborn (correlation heatmap, Figure 5.1) and Streamlit (dashboard prepared for presenting the analysis).

A.2 Analysis Parameters

Study period: 1 January 2020 to 1 January 2025 (five years of daily data). Annualisation: 252 trading days. Risk free rate: 4% per annum, held constant. Monte Carlo: 10,000 randomly weighted portfolios, seed np.random.seed(18). Rolling volatility window: 30 trading days.

Table A.1: Summary of Key Metrics

Metric	Equal Weight Portfolio	Maximum Sharpe Portfolio	Minimum Volatility Portfolio
Expected Annual Return	23.78%	29.53%	20.44%
Annualised Volatility	21.40%	22.60%	18.55%
Sharpe Ratio	0.92	1.13	0.89
Maximum Drawdown	33.24%	n/a	n/a
Average Rolling Volatility (30 day)	18.10%	n/a	n/a

A.3 Author Information

Author: Rasheedat Seidu. Project: Quant Foundations (Project 1). This report was prepared from the author's original Jupyter notebook analysis. All calculations, figures, tables and results were generated from that analysis.